

Letter

Space charge limited current in organic materials with free and trapped charges

Zhiwei Miao, Zhendong Ge, Dawei Gu, Lei Wang, Tianyou Zhang^{*}

Department of physics, Nanjing Tech University, Nanjing 211816, PR China

ARTICLE INFO

Keywords:

Exponentially distributed traps
Current density
Free and the trapped charges
Organic semiconductors

ABSTRACT

The space charge limited current of amorphous organic semiconductors with an exponential density of states (EDOS-SCLC model) resembles the same dependence of bias with the trap limited current (TLC model) case. The similarity of these two current density formulas may cause an overestimation of the influence of the free and trapped charges on current density when interpreting experimental data. We promote a transparent model incorporating both the influences of the free and the trapped charges on the current density. The model can be simplified into the EDOS-SCLC model and the TLC model in limiting conditions. When the influences of the free and the trapped charges are comparable, a transition of the slope of the current-voltage curve appears. When compared with experimental results, satisfactory agreement is obtained. What's more, the influence of dilution of narrow band materials into wide band matrix on the current density is discussed.

Introduction

Over the last few decades organic semiconductors have gained considerable interest due to their potential applications in Organic Light-Emitting Diodes (OLEDs), Organic Photovoltaics (OPV) and Organic photo-detectors (OPDs). Understanding the charge transport in organic semiconductors and the availability of physical-based models is important to improve the device performances and synthesize better materials.

Early study indicates that for some organic semiconductors at low charge concentration, the mobility is essentially constant and depends on temperature only. Hence, the classic Mott-Gurney model [1,2] $J = (9/8)\epsilon_0\epsilon_r\mu_p(V^2/L^3)$ holds. However, for most of the organic materials, mobility may not be a constant but a function of carrier concentration and of the electric field. What's more, for some materials the single carrier devices, mostly for minority carrier, suffer heavily from the influence of trapped charges. In 1962, P. Mark and W. Helfrich, proposed a trap-limited current (TLC) model for single carrier devices with an exponentially distributed traps [3,4]. In the TLC model, the distribution parameter T_c could be obtained from the slope of current density. And this model has been widely used and generalized to interpret experimental data. In 1996, P. W. M. Blom et al. measured the electron and hole current of PPV [5] and found that the electron current could be described with the TLC model. In 2007, to overcome the contradiction

between prediction of TLC model and the temperature dependence of the electron current in PPV derivatives, the exponential distribution of mobile carriers was extended to the Gaussian distribution by Mandoc et al. [6]. Furthermore, H. T. Nicolai et al. extended the exponential trap density of states (EDOS) to the Gaussian trap DOS [7]. Based on the extended model, the authors reinterpreted the low electron currents and steep voltage dependence in some PPV derivatives. Interestingly, in 2016, D. Abbaszadeh et al. conducted an experiment to dilute PPV derivatives into wide band gap polymers to heal the trap effect and to enhance the electron current [8]. And the experimental results perfectly validated the predictions of the TLC model.

On the other hand, the energetic disorder of amorphous organic materials leads to a Gaussian distribution of density of states which usually be replaced by an EDOS for simplification. For the specific case of EDOS, Vissenberg and Matters [9,10] advanced a percolation mobility model (VM mobility). In this model, the mobility is expressed as a function of carrier density:

$$\mu = a\rho^{b-1}, \quad (1)$$

where a is a material- and temperature (T)-dependent pref-factor, and $b = T_0/T$, with T_0 the width of the DOS. And in 2006, based on the VM mobility, B. Ramachandran et al. derived a Space-charge-limited current (SCLC) model for the case of EDOS (EDOS-SCLC) [11]. In 2010, F. Torricelli et al. got exactly the same current expression and the influence

^{*} Corresponding author.

E-mail address: tyzhang@njtech.edu.cn (T. Zhang).

of electric field on mobility was also discussed [12]. Interestingly, the TLC model and EDOS-SCLC model, though considering different situations, possess entirely the same form of dependence on bias. The only difference between these two models is the parameter T_c in TLC has been replaced by T_0 for the case of EDOS. This fact might cause difficulties in explaining experimental results. We notice that when deriving these two models, the free carriers and the trapped carriers have been omitted respectively. However, both free and trapped carriers should have effect on the current density.

In this work, we consider both the influences of the free charges and of the trapped charges on the current density. When both free and trapped carriers are introduced into the Poisson's equation, the current density expression could not be worked out analytically. Numerical calculation suggests that the slope of current density will depend both on the free carriers and on the trapped carriers, T_0 and T_c , when the contributions to the Poisson's equation are comparable. This result indicates a strong overestimate of T_c and T_0 in special cases. The new model is consistent with experimental results in literature.

TLC model and EDOS-SCLC model

To separately study hole current and electron current, single carrier devices should be fabricated with at least one Ohmic contact between the organic material and the electrodes. The electron current and hole current are equivalent for the TLC model and EDOS-SCLC model. Herein, we take hole-only devices as an example. Under low charge concentration, if we ignore the diffusion current, the Mott–Gurney expression of current with a constant mobility μ_p is [1]:

$$J = \frac{9}{8} \epsilon_0 \epsilon_r \mu_p \frac{V^2}{L^3}, \quad (2)$$

with J the current density, $\epsilon_0 \epsilon_r$ the permittivity of the device, μ_p the hole mobility, V the applied voltage, and L the thickness of the device. In this current expression, J is directly proportional to the V^2 . Hence the mobility could be directly obtained by measuring the current density slope.

For some organic materials, there may be a distribution of traps that influence the current density severely. These traps may be introduced during device fabrication or intrinsically from the tail of density of states. What's more, to fulfill particular functions, device may be engineered as host-guest system or bulk hetero-junctions in which traps can not be ignored. Assuming that the hole mobility is field-independent, P. Mark and W. Helfrich derived the TLC model for the case of exponential trap distribution [3,13]. The derivation process is outlined as follows.

The density of hole trapping states per unit energy range $h(E)$ above the valence band is described by the distribution:

$$h(E) = \frac{H}{kT_c} \exp\left(-\frac{E}{kT_c}\right), \quad (3)$$

with E the energy measured upward from the top of the valence band, H the total trap density, and T_c the characteristic constant of the distribution. Then the density of positive trapped space-charge $p(x)$ is:

$$p(x) = \int_{E_{FP}(x)}^{\infty} \frac{H}{kT_c} e^{-\frac{E}{kT_c}} dE, \quad (4)$$

with $E_{FP}(x)$ the quasi-Fermi level for holes. The density of positive free space-charge $p_f(x)$ can be derived as:

$$p_f(x) = \int_{E_{FP}(x)}^{\infty} \frac{N_0}{kT} e^{-\frac{E}{kT}} dE, \quad (5)$$

with N_0 the effective density of states in the valence band.

From Eqs. (4) and (5), the relation between the density of free

charges and the density of trapped charges is obtained as:

$$p_f(x) = N_0 \left[\frac{p(x)}{H} \right]^c, \quad (6)$$

where $c = T_c/T$.

Combined with the Poisson's equation and the drift current density:

$$\frac{dF(x)}{dx} = \frac{qp(x)}{\epsilon_r \epsilon_0}, \quad (7)$$

$$J = q\mu_p p_f(x) F(x), \quad (8)$$

with F the electric field strength, an equation for electric field F can be obtained:

$$\frac{dF}{dx} = D[F(x)]^{-\frac{1}{c}}, \quad (9)$$

where

$$D = \frac{H}{\epsilon_r \epsilon_0} q^{\frac{c+1}{c}} \left(\frac{J}{N_0 \mu_p} \right)^{\frac{1}{c}}. \quad (10)$$

Using the boundary condition $F(0)=0$, the expression of F could be integrated from Eq. (9). To obtain the applied voltage V as a function of J , the electric field is integrated from $x = 0$ to $x = L$. After rearrangement, we could get the TLC formula:

$$J = \frac{N_0}{H^c} \mu_p q^{1-c} \left(\frac{\epsilon_r \epsilon_0 c}{c+1} \right)^c \left(\frac{2c+1}{c+1} \right)^{c+1} \frac{V^{c+1}}{L^{2c+1}}. \quad (11)$$

According to this current density expression, parameter T_c could be obtained from the slope of the J - V plot.

It should be noted that, within the TLC model, the carrier mobility is assumed as a constant. Actually, HOMO energy level is not a single energy level, but distributed with complex shapes which influence the mobility directly. Following the study of Vissenberg and Matters [9], at low carrier densities, the transport properties are determined by the tail of an exponential density of the localized states. And the VM mobility is written as:

$$\mu_{VM} = \mu(T, P, F=0) = \mu_0(T) p_f(x)^{\frac{T_0}{T}-1} = \frac{\sigma_0}{q} \left[\frac{\left(\frac{T_0}{T} \right)^4 \sin\left(\pi \frac{T}{T_0} \right)}{(2\alpha)^3 B_c} \right]^{\frac{T_0}{T}} p_f(x)^{\frac{T_0}{T}-1}, \quad (12)$$

where σ_0 is a prefactor for the conductivity, α^{-1} is the effective overlap parameter between localized states, T_0 is a parameter that depends on the material disorder, and B_c is the critical number for the onset of percolation. Eq. (12) is a specific version of Eq. (1).

Using the VM mobility, Torricelli et al. successfully derived the current density by solving the Eq. (8) and Poisson's equation [12]:

$$\frac{dF(x)}{dx} = \frac{qp_f(x)}{\epsilon_r \epsilon_0}. \quad (13)$$

After integration the current density expression is written as:

$$J = q^{1-b} \mu_0(T) \left(\frac{\epsilon_0 \epsilon_r b}{b+1} \right)^b \left(\frac{2b+1}{b+1} \right)^{b+1} \frac{V^{b+1}}{L^{2b+1}}, \quad (14)$$

here $b = T_0/T$. Similarly, T_0 can be deduced from the slope of the J - V plot.

However, comparing these above two models, we can find that these two models have exactly the same form, though from different starting points. This causes confusion for determining T_0 and T_c through the slope of the J - V curve. Comparing Eqs. (7) and (13), we can see that the charges considered in the Poisson's equation by these two models are

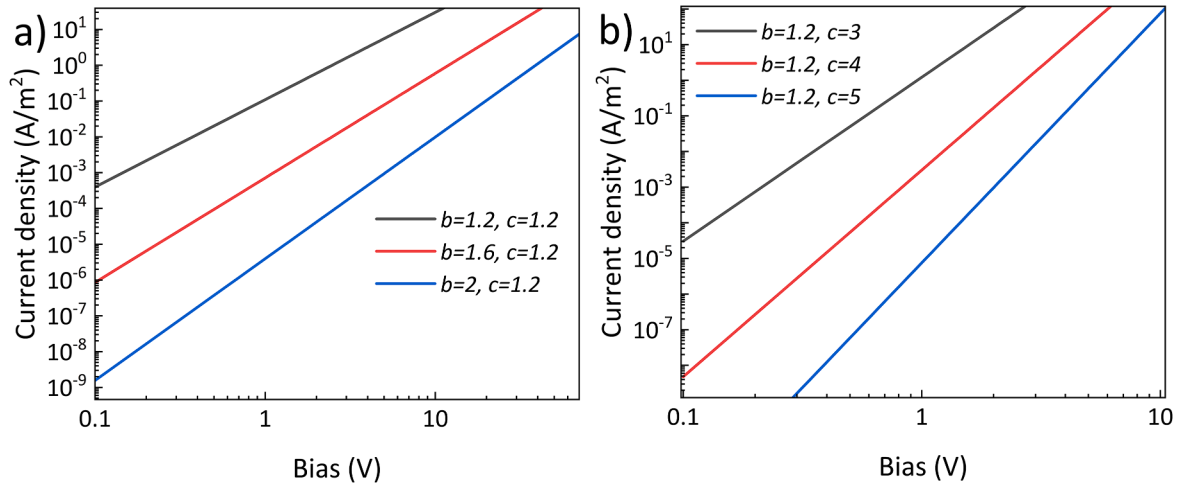


Fig. 1. The J - V characteristics [Eq. (15)] with selected b or c values. Other parameters are $\sigma_0=700$ S/m, $\alpha^{-1} = 0.12$ nm, $B_c=2.8$, $N_0=10^{30} \text{ m}^{-3}$, $L = 220$ nm, influence of parameter b ($H = 10^{28} \text{ m}^{-3}$). b), influence of parameter c ($H = 10^{24} \text{ m}^{-3}$).

different, one is the trapped charges $p(x)$ and another is the free charges $p_f(x)$. If both charges exist, whether the slope of the J - V curve is used for determining T_0 or T_c , the results will be overestimated. Therefore, an entire consideration is imperative.

Current model with free and trapped charges

In this section, we examine the effect of both free and trapped charges on the SCLC model. Before an entire consideration, we first introduce the VM mobility into the TLC model. Combining Eqs. (6)–(8) and (12), similar to the procedure of TLC and considering only trapped charges in Poisson's equation, a new current density expression is written as:

$$J = \frac{N_0^b}{H^{bc}} \mu_0(T) q^{1-bc} \left(\frac{\epsilon_r \epsilon_0 b c}{bc+1} \right)^{bc} \left(\frac{2bc+1}{bc+1} \right)^{bc+1} \frac{V^{bc+1}}{L^{2bc+1}}. \quad (15)$$

On one hand, this current density expression can be viewed as EDOS-SCLC model with traps. At the limit $c = 1$, Eq. (15) becomes Eq. (14). On the other hand, it can also be viewed as TLC model with non-constant mobility. In the limit condition $b = 1$, Eq. (15) is exactly the same of TLC model, i.e., Eq. (11). Fig. 1 shows the influence of parameters b and c on the slope of J - V plots described by Eq. (15). To the best of our knowledge, the values of T_c and T_0 of the materials given in the literature is limited to about 1600 K and 550 K, so we focus on the range of 0–5 and 0–2 for c and b respectively.

In Fig. 1a), with fixed c , the slope of current density increases as a

result of an increase of b value. Similarly, in Fig. 1b), with fixed b value, as the value of c increases, the slope of current density increases. For both cases, the slope of current density is expressed as $bc+1$. As the values of b and c increase, the magnitude of current density depresses significantly. It should be noted that, Eq. (15) has a fatal limitation. When the value of H goes to zero, that is no trap states involves, the situation should be simplified to the case of EDOS-SCLC, as described by Eq. (14). However, from Eq. (15), the magnitude of current density goes to infinite if H approaching zero. This is because the TLC model only takes into account the trapped charges and excludes the free charges. Therefore, a full version of Poisson's equation is imperative:

$$\frac{dF(x)}{dx} = \frac{qp_f(x) + qp(x)}{\epsilon_r \epsilon_0}. \quad (16)$$

By inserting Eqs. (8) and (9) into Eq. (16), we have:

$$\frac{dF}{dx} = D[F(x)]^{-\frac{1}{bc}} + A[F(x)]^{-\frac{1}{b}}, \quad (17)$$

where

$$D = \frac{H}{\epsilon_0 \epsilon_r N_0^{\frac{1}{bc}}} \left[\frac{J}{\mu_0(T) q} \right]^{\frac{1}{bc}} q \quad (18)$$

and

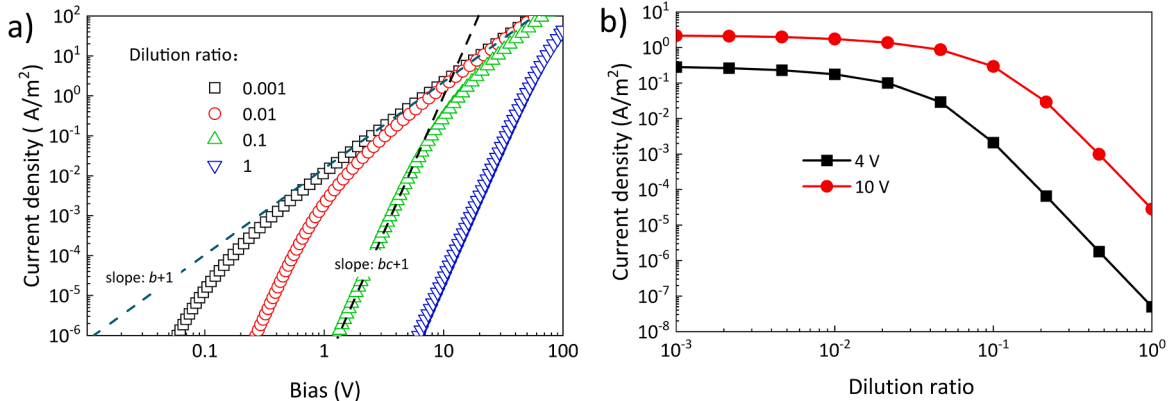


Fig. 2. a), The J - V characteristics at different dilution ratio [Eq. (16)]. b), The magnitudes of current density as a function of dilution ratio under selected biases. Other parameters are $\sigma_0=1000$ S/m, $\alpha^{-1} = 0.12$ nm, $B_c=2.8$, $N_0=10^{26} \text{ m}^{-3}$, $H = 10^{24} \text{ m}^{-3}$, $L = 270$ nm, $T_0=350$ K and $T_c=1500$ K.

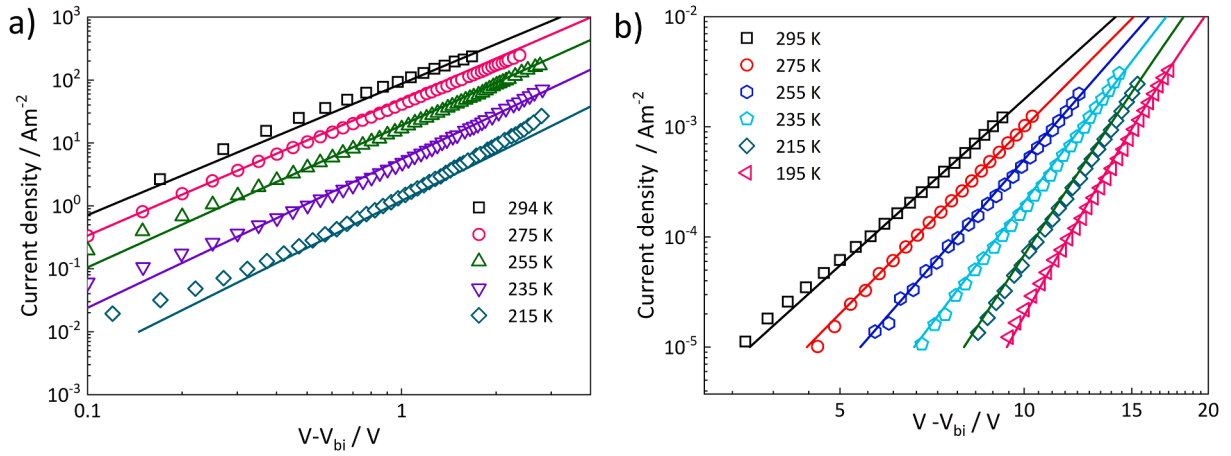


Fig. 3. Temperature dependence of the current for two different organic semiconductors. a), Temperature dependent current density vs voltage characteristics of a P3HT hole-only diode, with thickness 95 nm [14]. Symbols are experimental data from reference 14, solid lines are calculated results from Eq. (16). b), Temperature dependence of the electron current for MEH-PPV, with the thickness 270 nm [7]. Symbols are experimental data from references 6 and 7, solid lines are calculated results from Eq. (16).

$$A = \frac{1}{\epsilon_0 \epsilon_r} \left[\frac{J}{\mu_0(T) q} \right]^{\frac{1}{b}} q. \quad (19)$$

The electric field distribution and the current density could not be integrated analytically from Eq. (17). Thus we have to solve it numerically. Fig. 2a) illustrates several current density curves calculated from Eq. (17) with selected b and c values. It is obvious that, different from pure VM mobility and TLC model, the slope of the current density would decrease as an increase of voltage at certain b and c values. This phenomenon can be understood by comparing coefficients D and A in Eq. (17). If the value of D is much greater than that of A , then the model simplified to the TLC model. While if $A \gg D$, the model would be simplified as SCLC with VM mobility. If the magnitudes of A and D are comparable, the slope of current will shift as the voltage increase. As indicated in Fig. 2a) by the dashed lines, the slope of the current shift from $bc+1$ to $b+1$.

The ratio between the magnitude of D and A could be written as:

$$\frac{D}{A} = \frac{H}{N_0^{\frac{1}{c}}} \left(\frac{J}{aq} \right)^{\frac{1}{bc} - \frac{1}{b}}. \quad (20)$$

As expressed in Eq. (20), parameters a , b , c , H , N_0 and the magnitude of current density could influence the $D:A$ ratio therefore the slope of the current density. As shown in Fig. 2a), with the same current, the slope of the curve gradually changed from $bc+1$ to $b+1$ with the dilution ratio decreasing. And this means that our model changes from the TLC model to the EDOS-SCLC model. Meanwhile, a transition between these two models can be observed from Fig. 2a). From Fig. 2b), we can find that the current increases with the dilution ratio decreasing. And the current at dilution ratio range of 10^{-2} – 1 increases faster than the one at dilution ratio 10^{-3} – 10^{-2} . When the dilution ratio is low enough, the value of the current will not be virtually affected.

An interesting phenomenon could be predicted from the TLC model with exponentially distributed traps, the current scales with N_0/H^c [8]. A manipulation of N_0/H could be achieved by controlling the doping concentrations of one material into a matrix material. Using this strategy D. Abbaszadeh et al. successfully enhanced the LED efficiency by eliminating the trapping effect of electron [8]. Our model inherits this dilute character as it is a generalization of the TLC model.

To test the robustness of our model when interpreting experimental data, we use our model to fit different experimental results. As discussed above, if $H = 0$, our model simplified as pure EDOS-SCLC model. Therefore, we first fit our model with the current of a P3HT hole-only diode. The J - V curves under different temperatures for P3HT, though

with different curvatures but all almost line with single slopes. The slopes are great than 2 but not very large (the TLC case is rule out), suggesting that the mobility is not constant but maybe a function of carrier density or electric field. The electric field influence on mobility under low temperature (215 K) though becomes obvious but not a dominant factor than carrier density. In other word, the J - V curves should be governed by the carrier density dependence of mobility. This mechanism is exactly one limit case of our model. As shown in Fig. 3a), in the temperature range 255–294 K, acceptable agreement between numerically calculated results and experimental results achieved. The values of parameters are as: $T_0=320$ K, $\sigma_0=3.5 \times 10^4$ S/m, $\alpha^{-1} = 0.16$ nm and $B_c=2.7$ [15,16].

On the other hand, if $b = 1$ and a constant mobility is assumed, our model can be simplified as the TLC model [17]. It should be noted that $\mu_0(T)$ according to Eq. (12) will be 0 when $b = 1$, since the value of $\sin(\pi T/T_0)$ is 0. According to the percolation theory of VM mobility, the $\sin(\pi T/T_0)$ stems from an analytical charge density expression which holds only for $T < T_0$. As a matter of fact, if $b = 1$ ($T_0=T$) the charge density should be replaced and expressed exactly the form: $n = N_0 e^{E_F/kT} \ln(1 + e^{E_F/kT})$. With the same percolation procedure, a non zero mobility can be obtained. We apply our model to fit the electron current of a PPV derivative. As shown in Fig. 3b), the electron transport of MEH-PPV can be well described with parameters T_0 , T_c , σ_0 , α^{-1} , namely $T_0=300$ K, $T_c=1160$ K, $\sigma_0=3 \times 10^3$ S/m, and $\alpha^{-1} = 0.14$ nm at room temperature. However, as demonstrated by M. M. Mandoc et al. [6], the TLC model (with constant total trap sites) predict strong temperature dependence of current density. To given a consistent description for the entire range of temperature measured, a Gaussian DOS of conduction band and a symmetric exponential trap DOS should be incorporated. As a result, the total density of traps H in Eq. (16) should be adapted as $N_{t(\text{eff})} = N_t \exp[(E_{tc} - E_a)/kT]$, with E_{tc} the center of symmetric exponential trap DOS, $E_a = \sigma^2/kT$ (σ the variable of Gaussian transport DOS). Therefore, at low temperatures, a roughly exponential decrease values of $N_{t(\text{eff})}$ are used, at 295 K, 275 K, 255 K, 235 K and 215 K, the values of $N_{t(\text{eff})}$ used are: $4.4 \times 10^{24} \text{ m}^{-3}$, $3.5 \times 10^{24} \text{ m}^{-3}$, $2.2 \times 10^{24} \text{ m}^{-3}$, $1.4 \times 10^{24} \text{ m}^{-3}$, $8.8 \times 10^{23} \text{ m}^{-3}$, and $5.7 \times 10^{23} \text{ m}^{-3}$, respectively. A temperature dependent equilibrium between the trapped and free charges indicating that the amount of free and trapped charges maybe influenced by temperature. This equilibrium should not be DOS dependent. This is in consistent with the discussion of Mandoc et al. and Nicolai et al. [6,7]. Each J - V curve exhibits a single slope in Fig. 3b), indicating that the influence of trapped charges on the current density is the leading mechanism. While the diffusion current and a electric field dependent mobility considered in references have only weak influences on the

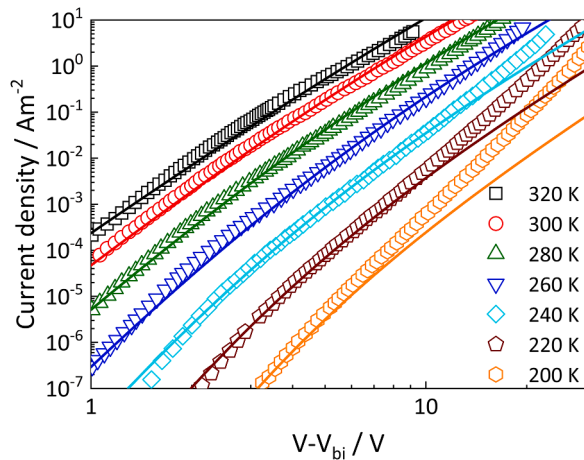


Fig. 4. The device temperature dependence of the J–V characteristics of an Alq3/Ca device with an Alq3 thickness of 300 nm. Symbols are data from Ref. 4. Lines are fits to Eq. (16). Other parameters are $\sigma_0 = 2 \times 10^7$ S/m, $\alpha^{-1} = 0.3$ nm, $B_c = 2.7$, $N_0 = 1.56 \times 10^{27}$ m $^{-3}$, $L = 270$ nm, $T_0 = T_c = 650$ K, $N_{t(\text{eff})} = 1.1 \times 10^{25}$ m $^{-3}$, 6.0×10^{24} m $^{-3}$, 3.6×10^{24} m $^{-3}$, 2.3×10^{24} m $^{-3}$, 1.4×10^{24} m $^{-3}$, 9.0×10^{23} m $^{-3}$, 6.0×10^{23} m $^{-3}$, at temperatures 320 K, 300 K, 280 K, 260 K, 240 K, 220 K, 200 K respectively.

current density [6,7].

Finally, we fit a more complicated results with our model. As shown in Fig. 4, the current density in an electron only diode based on Alq3 at different temperatures exhibit a decrease of its slope. This result have been theoretically explained by Brütting et al. [4]. with a TLC model with an electric field dependent mobility. Later in 2007, Limketkai et al., gave another explanation with a percolation mobility model as a function of both electric field and carrier density [18]. The original experimental results were measured down to 100 K. However, our model do not consider the influence of the electric field on mobility [19,20]. From Fig. 4 we can see that at low bias our model fit the experimental results very well. Parameters used are $\sigma_0 = 2 \times 10^7$ S/m, $\alpha^{-1} = 0.3$ nm, $B_c = 2.7$, $N_0 = 1.56 \times 10^{27}$ m $^{-3}$, $L = 270$ nm, $T_0 = T_c = 650$ K, $N_{t(\text{eff})} = 1.1 \times 10^{25}$ m $^{-3}$, 6.0×10^{24} m $^{-3}$, 3.6×10^{24} m $^{-3}$, 2.3×10^{24} m $^{-3}$, 1.4×10^{24} m $^{-3}$, 9.0×10^{23} m $^{-3}$, 6.0×10^{23} m $^{-3}$, at temperatures 320 K, 300 K, 280 K, 260 K, 240 K, 220 K, 200 K respectively. As the temperature decreases, the slope of current under different temperature increases heavily. And our calculated results deviate from the experimental data. Under temperature lower than 200 K, our model could not fit the curves any more (not shown). However, our explanation is different from the previous two. In our calculations, at low bias, the slope depends on both free and trapped charges, as $bc+1$. At high bias, the dependence of the free charges dominates, therefore the slope of current density decreases to $b+1$. What's more, in our model because both free and trapped charges could affect the slope of the current density, the fitted values of T_c and T_0 will be smaller than the one from pure VM SCLC and pure TLC models.

Further considerations should be noted for our model. First, for many organic materials the DOS distributed as a Gaussian function. For the case of Gaussian DOS, the mobility will be constant at low carrier density [21]. Second, as shown in our simulation of the current density of Alq3 based devices, the influence of electric field on carrier mobility should be incorporated in the model. What's more, carriers in the EDOS may serve as both trapped and free charges. These aspects deserve detailed discussion.

Conclusion

In conclusion, a model considering both the influence of the free and trapped charges on the current density is presented. Parameters b and c corresponding to the distributions of the free and trapped charges can influence the slope of J–V curves at low voltages, while at relative high

voltages the influence of parameter b becomes dominant. Numerical calculation derived from this model is validated with experimental data collected from different materials in a wide range of operating conditions. It is more appropriate to apply this model to explain the carrier current density than to use the TLC model alone or the SCLC model alone.

CRediT authorship contribution statement

Zhiwei Miao: Writing – review & editing, Data curation. **Zhendong Ge:** Project administration. **Dawei Gu:** Investigation. **Lei Wang:** Data curation. **Tianyou Zhang:** Supervision.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (No. 62275120) and by the Natural Science Fund of Colleges and Universities in Jiangsu Province (No. 20KJB140015).

References

- [1] N.F. Mott, R.W. Gurney, *Electronic Processes in Ionic Crystals*, Oxford University Press, 1940.
- [2] N.B. Kotadiya, H. Lu, A. Mondal, Y. Ie, D. Andrienko, P.W.M. Blom, G.A. H. Wetzelaer, Universal strategy for Ohmic hole injection into organic semiconductors with high ionization energies, *Nat. Mater.* 17 (2018) 329–334.
- [3] P. Mark, W. Helfrich, Space-charge-limited currents in organic crystals, *J. Appl. Phys.* 33 (1962) 205–215.
- [4] W. Brütting, S. Berleb, A.G. Mückel, Device physics of organic light-emitting diodes based on molecular materials, *Org. Electron.* 2 (2001) 1–36.
- [5] P.W.M. Blom, M.J.M. de Jong, J.J.M. Vleggaar, Electron and hole transport in poly (p-phenylene vinylene) devices, *Appl. Phys. Lett.* 68 (1996) 3308–3310.
- [6] M.M. Mandoc, B. de Boer, G. Paasch, P.W.M. Blom, Trap-limited electron transport in disordered semiconducting polymers, *Phys. Rev. B* 75 (2007) 193202–193205.
- [7] H.T. Nicolai, M.M. Mandoc, P.W.M. Blom, Electron traps in semiconducting polymers: exponential versus Gaussian trap distribution, *Phys. Rev. B* 83 (2011) 195204–195208.
- [8] D. Abbaszadeh, A. Kunz, G.A. Wetzelaer, J.J. Michels, N.I. Crăciun, K. Koynov, I. Lieberwirth, P.W.M. Blom, Elimination of charge carrier trapping in diluted semiconductors, *Nat. Mater.* 15 (2016) 628–633.
- [9] M.C.J.M. Vissenberg, M. Matters, Theory of the field-effect mobility in amorphous organic transistors, *Phys. Rev. B* 57 (1998) 12964–12967.
- [10] F. Torricelli, L. Colalongo, Unified mobility model for disordered organic semiconductors, *IEEE Electron Device Lett.* 30 (2009) 1048–1050.
- [11] B. Ramachandran, H.G.A. Huizing, R. Coehoorn, Charge transport in metal/semiconductor/metal devices based on organic semiconductors with an exponential density of states, *Phys. Rev. B* 73 (2006) 233306–233309.
- [12] F. Torricelli, D. Zappa, L. Colalongo, Space-charge-limited current in organic light emitting diodes, *Appl. Phys. Lett.* 96 (2010) 113304–113306.
- [13] Y.B. Zhu, K. Zheng, X. Wu, L.K. Ang, Enhanced stability of filament-type resistive switching by interface engineering, *Sci Rep* 7 (2017) 43664–43670.
- [14] C. Tanase, E.J. Meijer, P.W.M. Blom, D.W. de Leeuw, Unification of the hole transport in polymeric field-effect transistors and light-emitting diodes, *Phys. Rev. Lett.* 91 (2003) 216601–216604.
- [15] G.E. Pike, C.H. Seager, Percolation and conductivity: a computer study. I, *Phys. Rev. B* 10 (1974) 1421–1434.
- [16] Vinay Ambegaokar, B.I. Halperin, J.S. Langer, Hopping conductivity in disordered systems, *Phys. Rev. B* 4 (1971) 2612–2620.
- [17] G. Paasch, S. Scheinert, Space charge layers in organic field-effect transistors with Gaussian or exponential semiconductor density of states, *J. Appl. Phys.* 101 (2007) 024514–024526.

- [18] B.N. Limketkai, P. Jadhav, M.A. Baldo, Electric-field-dependent percolation model of charge-carrier mobility in amorphous organic semiconductors, *Phys. Rev. B* 75 (2007) 113203–113206.
- [19] C. Tanase, P.W.M. Blom, D.M. de Leeuw, Origin of the enhanced space-charge-limited current in poly(p-phenylene vinylene), *Phys. Rev. B* 70 (2004) 193202–193205.
- [20] N.I. Crăciun, J. Wildeman, P.W.M. Blom, Universal arrhenius temperature activated charge transport in diodes from disordered organic semiconductors, *Phys. Rev. Lett.* 100 (2008) 056601–056605.
- [21] J.O. Oelerich, D. Huebner, S.D. Baranovskii, How to find out the density of states in disordered organic semiconductors, *Phys. Rev. Lett.* 108 (2012) 226403–226407.